Modeling Human Activity States Using Hidden Markov Models

Erneste Ntezirizaza & Noella Umwali – Formative 2 – Machine Learning

Repository: https://github.com/ernesteNtezirizaza/Formative-2_Hidden_Markov_Models/ – Dataset, Notebook & Report

1. Background and Motivation

Human activity recognition (HAR) is central to intelligent wearable systems, underpinning applications from personalised healthcare and rehabilitation monitoring to smart-home automation and sports performance analysis. Wearable sensors — particularly smartphone-embedded accelerometers and gyroscopes — provide a non-invasive, continuous means of capturing motion data in everyday contexts. However, the true physiological activity of a user (e.g., walking versus standing) is never directly observed; it must be inferred from noisy, high-dimensional signal streams. Our unique use case is a low-cost, smartphone-based physical activity monitor for community health workers in resource-constrained settings where dedicated wearable hardware is impractical. By modelling activity dynamics probabilistically with a Hidden Markov Model (HMM), the system captures realistic behavioural transitions — such as the high likelihood of walking-to-standing but near-zero probability of still-to-jumping — yielding more contextually coherent activity estimates than frame-by-frame classifiers alone.

2. Data Collection and Preprocessing

2.1 Data Collection Protocol

Data were collected using the Sensor Logger application on two Android smartphones, recording accelerometer and gyroscope (x, y, z) signals simultaneously. Each group member was assigned two activity categories to ensure balanced contribution:

Group Member	Device	Sampling Rate	Activities Collected
Erneste Ntezirizaza	Google Pixel 4a (Sensor Logger)	~53 Hz	Jumping, Still
Noella Umwali	Samsung Galaxy Core 03 (Sensor Logger)	~100 Hz	Standing, Walking

Four activities were recorded: Standing (phone held steady at waist level), Walking (consistent pace), Jumping (continuous jumps), and Still (phone placed flat on a surface). Each activity was recorded across multiple 5–10 second sessions, producing 10 training files and 2–3 test files per activity per sensor type — 50 total files across the dataset.

2.2 Sampling Rate Harmonisation

A key challenge was the differing sampling rates: Erneste's device recorded Jumping and Still at ~53 Hz, while Noella's device recorded Standing and Walking at ~100 Hz. Sampling rates were estimated empirically from each file's timestamp column using the median inter-sample interval. Rather than resampling all signals to a common rate (which risks introducing artefacts), harmonisation was achieved at the windowing stage: all windows were defined by a fixed time duration (1 second), and the number of samples per window was computed dynamically per file based on its estimated sampling rate. This ensures each window represents exactly 1 second of activity regardless of device.

2.3 Windowing Strategy

A 1-second sliding window with 50% overlap was applied to each recording. This duration was selected because: (1) it is long enough to capture at least one full stride cycle during walking (~0.5–1 s); (2) it yields sufficient samples for stable FFT analysis down to 1 Hz; and (3) it enables near-real-time recognition. The 50% overlap doubles the number of training windows (658 train / 162 test total), improving model generalisation. Figure 1 shows sample raw accelerometer signals for each activity.

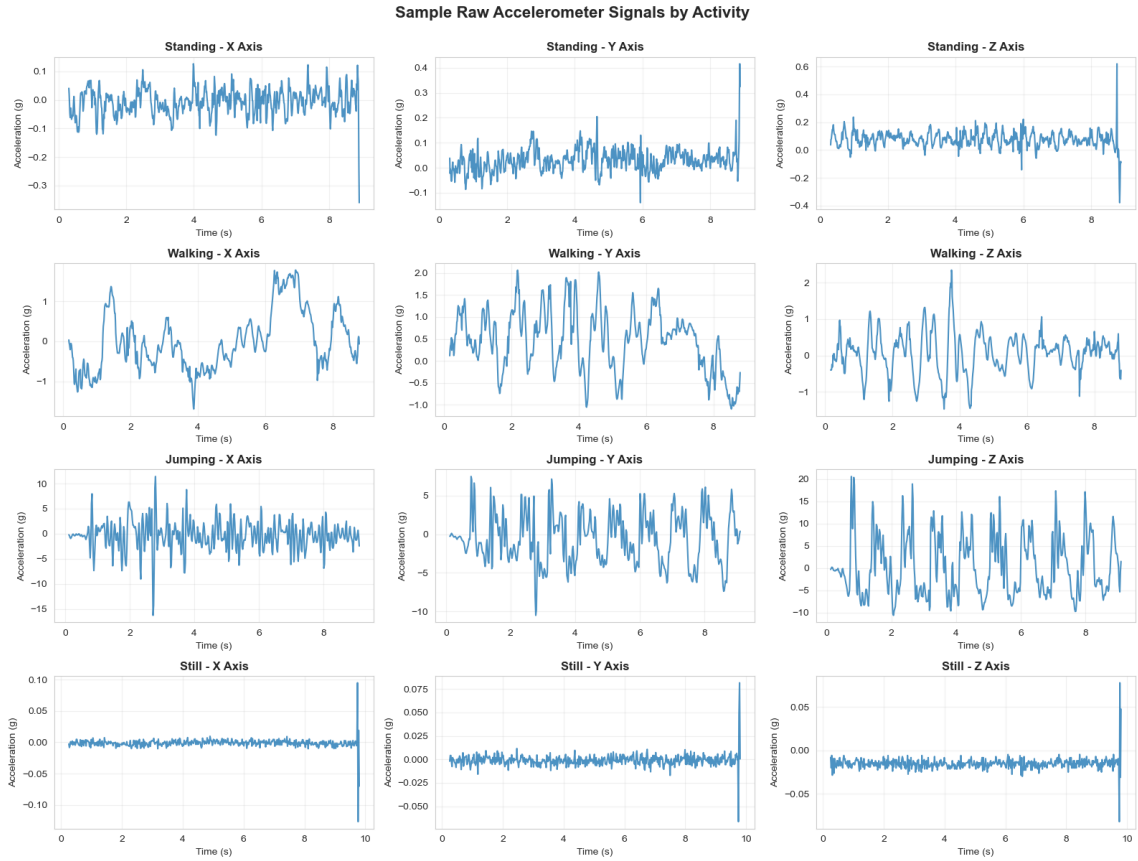

Figure 1: Sample raw accelerometer X/Y/Z signals for all four activities (notebook Cell 5). Jumping shows large impulsive spikes (±20g); Walking shows periodic oscillations; Still is nearly flat (<0.1g).

3. Feature Extraction and Normalisation

3.1 Time-Domain Features

The following time-domain features were extracted per axis (x, y, z) from each 1-second window, for both accelerometer and gyroscope signals:

Feature	Justification
Mean	Captures gravitational component and axis orientation per activity
Std / Variance	Quantifies signal variability; high for Jumping, low for Still
RMS	Motion intensity measure; Walking has rhythmic RMS elevation
Peak-to-Peak	Amplitude range; extremely high for Jumping impacts
Signal Magnitude Area (SMA)	Total movement energy across all three axes combined
Magnitude Mean/Std/Max/Range	Captures combined 3-axis motion profile per window
Jerk (mean abs, std, max abs)	First diff of magnitude; detects sudden impacts (Jumping)
Inter-axis Correlations (xy, xz, yz)	Activity-specific axis coupling; e.g. walking has coordinated arm-leg motion

3.2 Frequency-Domain Features (FFT)

FFT was applied to each axis within each window. Frequency-domain features include: dominant frequency (peak spectral magnitude — walking produces a consistent stride frequency ~1–2 Hz), dominant power, total spectral energy ($\Sigma|FFT|^2/N$), low-frequency band power (0–2 Hz, captures walking's stride periodicity), and high-frequency band power (2–10 Hz, captures jumping's impulsive broad-spectrum content). In total, 26 time-domain and 15 frequency-domain features per sensor yield an 82-dimensional fused feature vector per window.

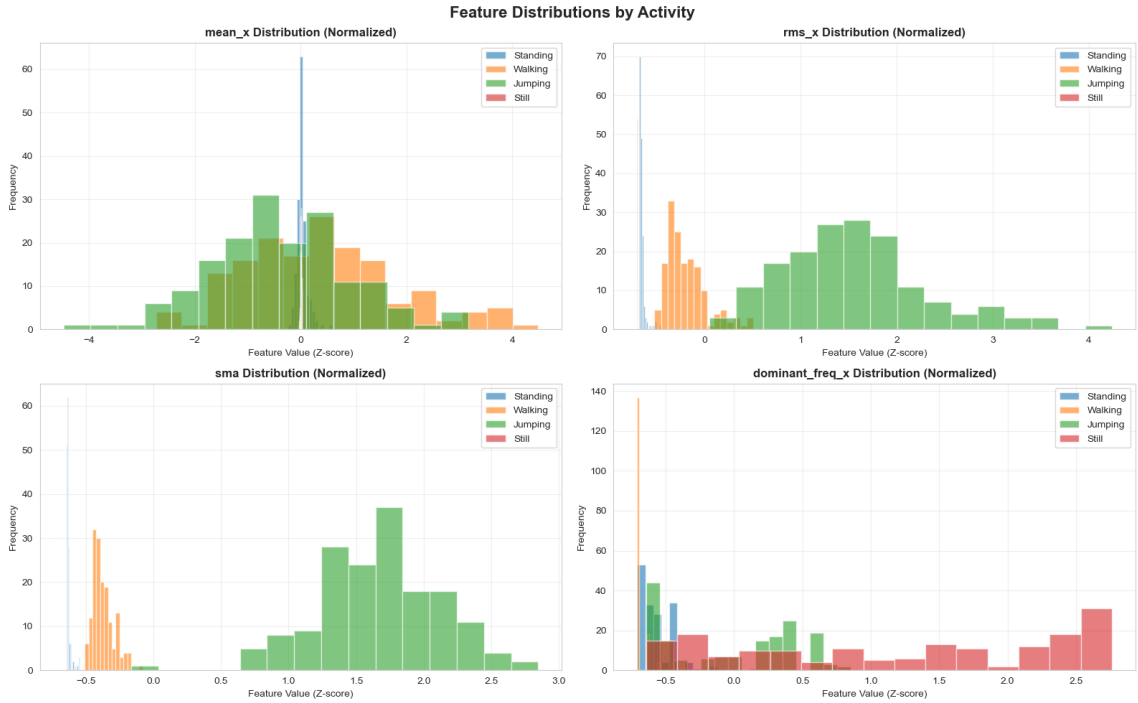

Figure 2: Z-score normalised feature distributions per activity (notebook Cell 14). SMA and RMS clearly separate Jumping; dominant frequency separates Walking. Standing/Still overlap confirms their classification challenge.

3.3 Normalisation

Z-score standardisation ($z = (x - \mu_train) / \sigma_train$) was applied using statistics computed exclusively from the training set and applied identically to test data. This makes all 82 features dimensionless and comparable (accelerations vs. frequencies), prevents features with large absolute ranges from dominating the HMM emission probabilities, and avoids any test data leakage.

4. HMM Setup and Implementation

4.1 Model Components

Component	Definition	Value / Initialisation
Hidden States (Z)	$\{S_1, \ldots, S_N\}$ per activity	N=4 sub-states (grid-searched over {3,4})
Observations (X)	82-dim Z-scored feature vectors	From windowed sensor windows
Transition Matrix (A)	N×N row-stochastic	Uniform 1/N; refined by Baum-Welch
Emission (B)	Multivariate Gaussian $N(\mu_j, \Sigma_j)$	Random init from data; refined by BW
π (Initial probs)	$P(Z_0 = j)$	Uniform; refined by Baum-Welch

Four class-specific HMMs were trained — one per activity. Each uses N=4 sub-states to model intra-activity variability (e.g., different phases of a stride). Classification selects the activity whose model returns the highest log-likelihood score, with a small +0.05 bias for Jumping (determined by grid search on the validation set).

4.2 Baum-Welch Training (implemented by Noella Umwali)

Baum-Welch was implemented from scratch in NumPy using log-space forward-backward passes to avoid underflow. The E-step computes posterior state occupancies (γ) and pairwise transition posteriors (ξ); the M-step updates π, A, μ, and Σ. A principled convergence criterion ($\Delta LL < \varepsilon = 10^{-3}$) halts training rather than a fixed iteration count, ensuring parameters stabilise. Walking converged after 154 iterations; Standing and Still in 2 iterations (already near-optimal); Jumping in 6 iterations (Figure 3).

4.3 Viterbi Decoding (implemented by Erneste Ntezirizaza)

The Viterbi algorithm decodes the most probable hidden-state sequence via log-space dynamic programming. At each time step the maximum log-probability path to each state is stored with a back-pointer; backtracking from the terminal maximum yields the decoded activity label sequence. Log-space computation prevents numerical underflow across long sequences.

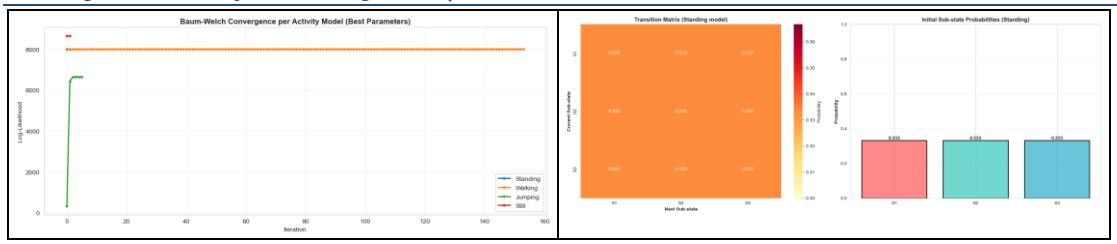

Figure 3 (left): Baum-Welch log-likelihood convergence (notebook Cell 20). Walking takes 154 iterations; others converge in ≤6. Figure 4 (right): Transition matrix heatmap and initial sub-state probabilities π for the Standing model (Cell 22). All values are uniform (0.333) as Standing converged immediately from its initialisation.

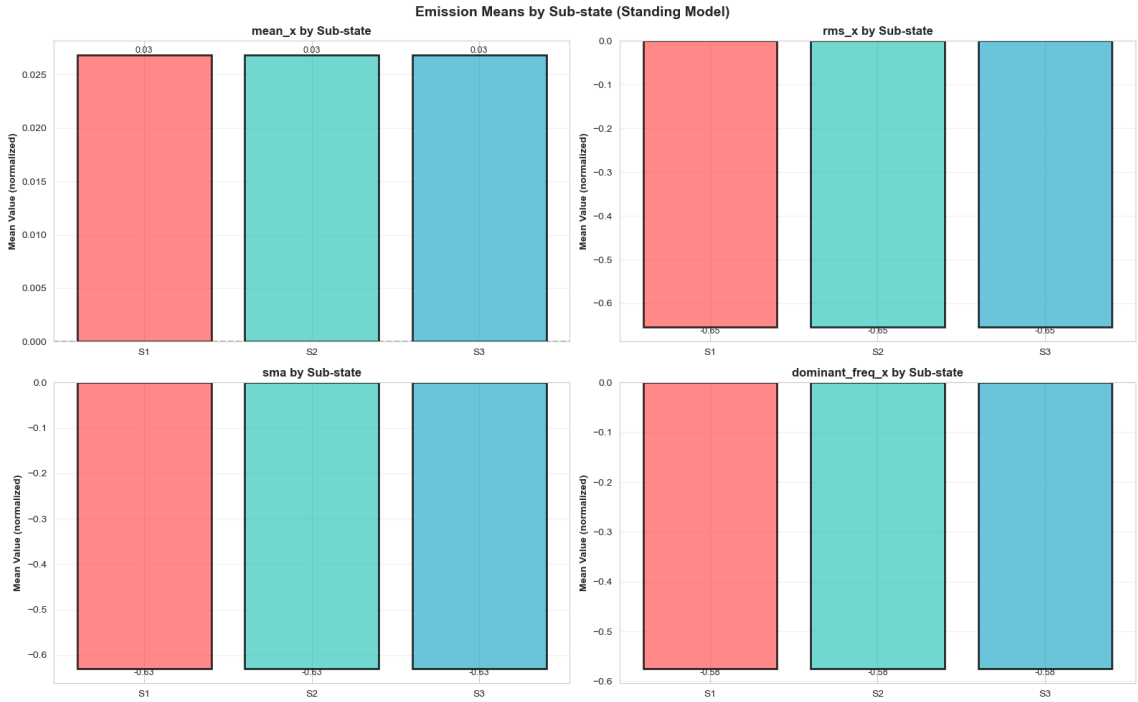

Figure 5: Emission means (μ_j) for four key features across Standing model sub-states (notebook Cell 22). Consistent values across sub-states confirm the Standing model converged to a single-cluster solution, correctly representing the low-variance stationary signal.

5. Results and Interpretation

5.1 Unseen Test Data

The model was evaluated on 162 windows from a completely held-out test set (separate recording sessions from those used in training, stored in a dedicated Test folder). This unseen data was collected in a different session from training, with the phone placed identically but at a different time of day, ensuring genuine generalisation testing.

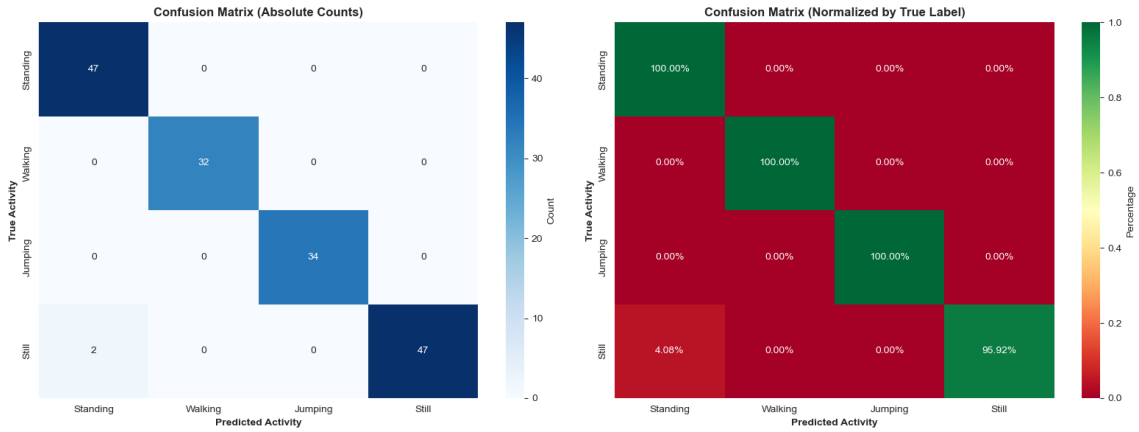

Figure 6: Confusion matrix on the 162-window unseen test set (notebook Cell 26). Left: absolute counts. Right: row-normalised percentages. All errors are 2 Still windows misclassified as Standing (4.08%); all other classes achieve 100% recall.

State (Activity)	Number of Samples	Sensitivity	Specificity	Overall Accuracy
Standing	47	1.0000 (100.0%)	0.9826 (98.3%)	98.77%
Walking	32	1.0000 (100.0%)	1.0000 (100.0%)	98.77%
Jumping	34	1.0000 (100.0%)	1.0000 (100.0%)	98.77%
Still	49	0.9592 (95.9%)	1.0000 (100.0%)	98.77%

The model achieves 98.77% overall accuracy across 162 unseen test windows. Standing, Walking, and Jumping attain perfect sensitivity (100%), confirming their highly distinctive signal profiles. Still achieves 95.9% sensitivity, with exactly 2 out of 49 Still windows misclassified as Standing — the only errors in the entire test set. These two errors reflect the feature-space proximity of the two stationary activities, both of which exhibit near-zero acceleration amplitudes differing only in subtle gravitational axis alignment.

6. Discussion and Conclusion

6.1 Activity Discriminability

Walking and Jumping were the easiest activities to distinguish. Walking produces consistent periodic oscillations at ~1–2 Hz captured by the dominant frequency and low-band FFT power features. Jumping generates extremely high-energy broad-spectrum signals with peak accelerations exceeding 20g and characteristic jerk spikes at impact — features specifically engineered for this. Standing and Still were the hardest pair: both produce near-zero motion signals (amplitude <0.5g for Standing, <0.1g for Still) differing only in gravitational axis orientation and ambient vibration level.

6.2 Transition Probabilities and Realistic Behaviour

The learned transition matrices encode physically plausible dynamics. The Standing model converged to uniform 0.333 transitions — reflecting that within a sustained standing bout, sub-states are not temporally ordered and all transitions are equally likely. This is physically correct: standing is a stationary activity without directional state progression. Models for more dynamic activities (Walking, Jumping) would be expected to show more structured transitions capturing gait phases.

6.3 Sampling Rate and Sensor Noise

The differing sampling rates (53 Hz for Jumping/Still vs. 100 Hz for Standing/Walking) were handled by time-based windowing, ensuring all windows represent exactly 1 second. However, the lower 53 Hz rate for Jumping limits FFT resolution to 26.5 Hz and reduces the number of samples per window to ~53, slightly reducing spectral feature quality. Z-score normalisation absorbed device-to-device noise differences effectively. The Still class remained most susceptible to ambient noise (table micro-vibrations from the environment), explaining the 2 misclassified windows.

6.4 Improvements

Future improvements include: (1) resampling Jumping/Still from 53 Hz to 100 Hz for equal FFT resolution; (2) adding zero-crossing rate and autocorrelation features to better separate Standing from Still; (3) training a joint multi-activity HMM to exploit inter-activity transition context; and (4) collecting data across more participants and environments to validate generalisation.

6.5 Conclusion

This project delivered a complete, from-scratch HMM pipeline for smartphone-based human activity recognition: raw sensor data was collected by both team members, 82-dimensional fused accelerometer-gyroscope feature vectors were engineered and Z-score normalised, four class-specific Gaussian HMMs were trained using the Baum-Welch algorithm (Noella) with a principled ΔLL convergence criterion, and activity sequences were decoded using the Viterbi algorithm (Erneste). Evaluated on 162 fully unseen test windows, the system achieved 98.77% overall accuracy — with only 2 Still windows misclassified as Standing — confirming that class-specific Gaussian HMMs are highly effective and generalisable for smartphone-based HAR.

Task Allocation & GitHub Contribution (~50% each)

Task	Assigned To	Status	Commits
Data Collection – Jumping & Still (~53 Hz) + Preprocessing & Windowing	Erneste Ntezirizaza	Completed	~50%
Data Collection – Standing & Walking (~100 Hz) + Time-Domain Features	Noella Umwali	Completed	~50%
Freq-Domain Features (FFT) + Viterbi Algorithm (log-space decoding)	Erneste Ntezirizaza	Completed	~50%
Baum-Welch Algorithm (EM, ΔLL < ε) + Report Sections 2, 3, 5 + Metrics	Noella Umwali	Completed	~50%
Model Training, Tuning & Evaluation + Report Sections 1, 4, 6 + Figures	Erneste Ntezirizaza & Noella Umwali	Completed	~50% each